

Project 7

Knowledge Exchange Platform – Coin-Based Marketplace System

Requirements Engineering & System Design Project

1. Project Overview

The Knowledge Exchange Platform is a structured system design project focused on building a regulated, coin-based peer-to-peer learning marketplace. Unlike an implementation-heavy project, this work concentrated on:

- Requirements engineering
- Economic system design
- Regulatory compliance modeling
- Database architecture design
- Release planning and prioritization

The platform allows users to act as learners and teachers, exchanging knowledge using an internal digital coin economy.

2. Core Problem

Traditional learning marketplaces:

- Depend on direct monetary transactions
- Lack structured internal economic models
- Often ignore compliance considerations
- Do not integrate trust and auditability at design level

The goal was to design a marketplace system that:

- Uses an internal coin-based economy

- Tracks all transactions securely
 - Ensures GDPR and AML compliance
 - Supports dual-role users (learner & teacher)
 - Maintains audit logs and traceability
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3. System Concept

The platform revolves around an internal currency system:

- New users receive initial coins.
- Learners spend coins to book sessions.
- Teachers earn coins.
- Coins can be topped up via payment gateway.
- Coins can be withdrawn after threshold.

This introduces:

- Internal economy control
 - Financial traceability
 - Fraud prevention possibilities
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4. Database Architecture

A relational data model was designed including:

- USER entity
- SESSION entity
- COIN_TRANSACTION ledger
- Skill categories
- Ratings
- Verification status

A permanent transaction ledger ensures:

- No transaction loss
- Auditability
- Financial integrity
- Compliance support

Relational constraints enforce:

- User-session mapping
 - Coin flow tracking
 - Transaction consistency
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5. Requirements Engineering Approach

The project applied formal requirements techniques:

- Functional requirements (PR-level)
- Domain-level requirements (DL-level)
- Non-functional requirements (performance, availability)
- Compliance requirements (GDPR, AML, KYC)

Quality attributes defined:

- 99.5% availability
- <2 second response time
- No transaction duplication
- High maintainability

This shows strong system-thinking maturity.

6. Prioritization & Release Planning

The project used:

- 100-Dollar Test prioritization
- Sprint-based release planning
- Incremental delivery roadmap

Release structure:

Release 1 → Core economy + booking

Release 2 → Trust & UX improvements

Release 3 → Compliance & payout maturity

This demonstrates structured product planning.

7. Compliance & Governance Considerations

The system includes:

- GDPR data export & deletion features
- AML and KYC modeling
- Financial traceability
- Audit logging

This elevates the system from simple marketplace to compliance-aware platform design.

8. Skills Demonstrated

- Requirements engineering
 - System architecture design
 - Economic model design
 - Compliance-aware system thinking
 - Database modeling
 - Product prioritization
 - Release strategy planning
 - Quality attribute definition
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9. Real-World Applicability

This design could be adapted for:

- EdTech platforms
 - Mentorship marketplaces
 - Token-based service platforms
 - Digital knowledge economies
 - Regulated online marketplaces
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This project strengthens your **system design + product architecture profile** rather than coding depth.

Next (final project):

RecklamRadar – Flutter Cross-Platform Application

Say:

NEXT

And I'll give brief structured documentation for that one.